Doctors vs Influencers:

Mapping Authority, Sentiment and Community in YouTube Health Discourse

COSC2671 Social Media and Network Analytics

Assignment 2 - Group Report

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1. Introduction

Social media's explosive expansion has completely altered how people obtain and disseminate health information. One of the biggest video platforms in the world, YouTube, features a wide variety of health-related content, from unverified wellness advice promoted by influencers to medically confirmed information offered by trained doctors. This divergence creates a significant public health concern: audiences may be equally exposed to scientifically grounded advice and potentially harmful misinformation.

This project examines how YouTube videos with medical professionals dispelling health myths differ from those made by wellness influencers endorsing alternative health advice in terms of comment behaviour, mood, linguistic patterns, and community structure. This study attempts to determine how viewers interact with each kind of information and whether comment networks represent unique communities that are in line with various health ideologies by using social media mining and network analysis approaches.

1.1. Research Question

The central research question guiding this project is:

"Can social network analysis of YouTube comment reply interactions reveal distinct user communities and influential actors in health misinformation discourse, and do these communities align with medical professional or wellness influencer content?"

1.2. Why this Problem Matters

Health misinformation on social media has real-world repercussions, such as vaccine hesitancy, risky food habits, and delays in obtaining competent medical care. Analysing comment communities reveals how disinformation spreads, who the prominent people are, and whether viewers build echo chambers around specific health views. This analysis is useful for public health academics, platform moderators, and policymakers.

1.3. Success Criteria

This analysis will be considered successful if it reaches the following:

- The study identifies and explains significant sentiment variations between doctor and influencer video comments.
- LDA topic modelling extracts distinct topics from video categories based on content type.
- A meaningful user interaction network is built using reply linkages, interpretable centrality measures, and communities.
- Community identification identifies user clusters around doctor or influencer material, indicating potential echo chambers.
- Identified and interpreted the network's most prominent users' roles.

2. Data Collection and Description

2.1. Data Source

Data was obtained from YouTube via the official YouTube Data API v3. YouTube was chosen as the data source because it provides a vast amount of health-related video from both medical professionals and wellness influencers, and its public comment system allows for network analysis of user interactions through reply threads.

2.2. Data Collection

Data was collected in two stages using a custom Python function (fetch YoutubeData) built around the YouTube Data API v3 client:

Stage 1 – Doctor/Debunking Videos :

Searched using 8 queries including "doctor debunks health influencer", "doctor reacts bad health advice", "doctor fact checks wellness influencer", "medical doctor debunks myths", and related terms.

Stage 2 — Influencer Videos:

Searched using 10 queries including "wellness influencer health tips", "carnivore diet influencer", "detox cleanse health influencer", "anti vaccine influencer", "raw milk benefits influencer", and related terms.

Videos were filtered to exclude YouTube Shorts (under 60 seconds) and low-engagement content, retaining only videos with 100 or more comments and 10,000 or more views. Up to 150 comments per video were fetched including top-level comments and replies.

The datasets collected from medical professional-focused queries and wellness influencer-focused queries were merged into a single unified dataset. Each video was assigned a label (doctor or influencer) to enable comparative analysis. This combined dataset allows for consistent analysis across both groups in terms of sentiment, topic modelling, and network interaction patterns.

2.3. Dataset Summary

- A total of 89 videos were retrieved across two collection stages , 60 from medical professional/debunking queries and 29 additional influencer videos , after deduplication and quality filtering.
- Up to 150 comments per video were collected, yielding 19,416 raw comments in total. After cleaning and filtering, 18,770 English comments were retained, providing a sizable and high-quality dataset for analysis.
- The gathered information includes attributes such as video ID, video title, channel name, view count, like count, comment text, comment author username, reply-to username, comment likes, and publication timestamp.
- Before being transformed into a structured format (CSV) for analysis, the dataset was first collected and stored in JSON format (mental_health_dataset.json), then flattened into a tabular DataFrame with one row per comment for further processing.

- Videos were labelled as doctor, influencer, or both based on keyword matching in the video title and channel name, enabling comparative analysis across sentiment, topic modelling, and network analysis stages.

Table 1: Dataset Summary and Preprocessing Steps

Metric	Value
Total videos collected	89
Doctor/debunking videos	50
Influencer videos	22
Both (mixed content) videos	17
Total comments (raw)	19,416
Comments after cleaning	18,770
Comments removed (cleaning)	646
Average comments per video	210.9
Min comments per video	99
Max comments per video	292
Average tokens per comment	13.6
Collection period	2026

2.4. Data fields

The following fields were used in the analysis:

Table 2: Description of Dataset Fields and their usage

Field	Description	Used For
video_id	Unique YouTube video identifier	Deduplication, grouping
title	Video title	Labelling, display
channel	Channel name	Labelling
view_count	Total video views	Filtering
like_count	Total video likes	Metadata
comment	Raw comment text	NLP analysis
author	Comment author username	Network nodes
reply_to	Username being replied to	Network edges
comment_likes	Likes on comment	Metadata
published_at	Comment timestamp	Metadata

2.5. Network Support

The reply_to field explicitly encodes user interaction relationships, allowing the creation of a directed reply network with nodes representing users and edges representing reply interactions. Of the 19,416 raw comments, 17,246 (88.8%) were top-level comments and 2,170 (11.2%) were replies, which served as the network's edges.

2.6. Limitations

- The YouTube Data API imposes quota limits, restricting the volume of data collectable per day.
- Keyword-based labelling of videos as doctor or influencer may misclassify edge cases where content overlaps.
- Comments are limited to English-language content (enforced via relevanceLanguage and ASCII filtering), which excludes non-English audiences.
- The dataset reflects a snapshot in time and may not represent long-term trends.
- The 'both' label category (17 videos) contains mixed content that is harder to interpret cleanly.

3. Data Preprocessing

3.1 Text Cleaning

The collected dataset was pre-processed to improve data quality and ensure consistency for analysis. The following steps were performed:

- To ensure consistency throughout the dataset, all comments were changed to lowercase.
- In order to get rid of unnecessary links that do not advance textual analysis, URLs were removed from the comments.
- Only significant textual information was kept after emojis and non-ASCII characters were eliminated.
- Only alphabetic text remained for examination after special characters and punctuation were eliminated.
- To prevent processing errors or useless data, empty comments were eliminated.
- Short comments (length $\leq$ 3) were eliminated because they do not offer enough information for analysis.
- ASCII checks were used in a filtering phase to eliminate comments that were not in English. Nonetheless, certain Romanised non-English comments such as Hindi written in English characters remain in the dataset since they nonetheless express insightful user ideas.
- There were 333 duplicates and these were removed to prevent bias and repetition in analysis.
- A cleaned version of the text was stored in a new column called clean_comment, which was used for further processing.

3.2 Tokenisation

In the next step, tokenisation was carried out using the TweetTokenizer from the NLTK package, which is specifically suited for social media material. This enables improved handling of colloquial language, acronyms, and user-generated material.

In addition to normal English stopwords, a special set of domain-specific stopwords was established, including phrases like "doctor," "influencer," and "youtube," to keep high-frequency but low-information words from dominating the study.

To increase the quality of textual characteristics, tokens that contained digits, URLs, or very short phrases were eliminated. The generated tokens represent significant words, which were then employed in sentiment analysis, topic modelling, and n-gram analysis.

There is a noticeable decrease in variability and extreme values in the boxplot comparison of comment lengths before and after preprocessing as shown in Figure 1. The dataset had a lot of lengthy comments and outliers before it was cleaned; these were probably caused by noise like URLs, repetitive language, and irrelevant content. The distribution becomes more compact after preprocessing, suggesting that noise has been successfully eliminated. There are still some outliers, though, which are real long-form user comments that add significant data to the analysis.

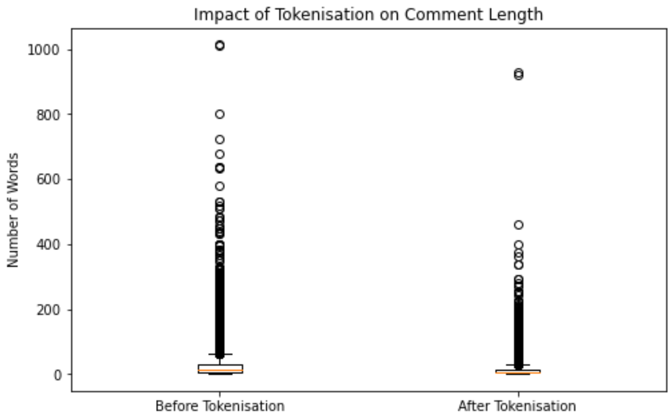

Figure 1: Boxplot Before and After Tokenisation

3.3 Video Labelling

Videos were labelled as belonging to one of three categories doctor, influencer, or both using a keyword scoring approach. By collecting instances of medical keywords ("doctor", "debunk", "fact check", "physician", "medical", "dermatologist", etc.) in the combined video title and channel name, a doctor score was determined. In a similar manner, influencer-related terms ("carnivore", "wellness", "detox", "raw milk", "supplement", "conspiracy", "naturopath", etc.) were used to create an influencer score. Videos were labelled "both" if the scores were equal, or whichever score was higher.

The label distribution across comments and unique videos is shown in Table 3 and visualised in Figure 2.

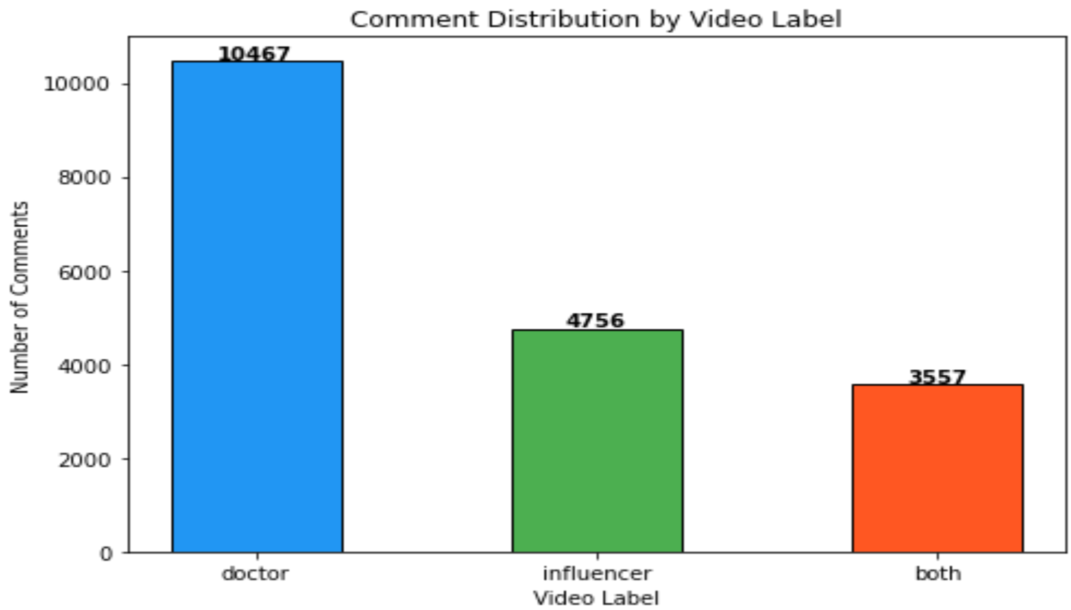

Figure 2: Comment Distribution by Video Label

Table 3: Distribution of Videos and Comments by Content Category

Label	Videos	Comments
Doctor	50	10,458
Influencer	22	4,756
Both	17	3,556
Total	89	18,770

Initial data investigation was carried out to determine the distribution of user involvement across videos. The histogram of comment counts demonstrates that most films receive a very consistent amount of involvement, ranging from 190 to 230 comments, indicating a balanced dataset with no extreme skew.

However, a bar chart of the top ten videos in Figure 2 shows that some videos receive substantially more participation. These videos are primarily associated with contentious or debate-provoking themes, such as food trends and influencer-based health advice.

These findings imply that, while overall involvement is pretty stable across the dataset, specialised themes result in higher levels of user participation. This demonstrates the impact of content type on audience engagement and lays the groundwork for additional investigation of sentiment, linguistic patterns, and interaction networks between medical professionals and wellness influencers.

4. Text Analysis

4.1 Unigram Analysis

As seen in Figure 3 and 4, the unigram analysis clearly distinguishes between the two content kinds. Doctor-led talks typically use broader, more general conversational language, reflecting explanatory and instructive content. In contrast, influencer-driven discussions are more focused and repetitious, focusing on specific health trends and dietary behaviours. This shows that medical experts encourage various health talks, whereas influencers tend to foster focused discourse on specific issues.

Notably, the top words in the "both" category include Portuguese/Spanish keywords ("de", "e", "que"), demonstrating the bilingual character of the carnivore diet group and international viewers watching debunking information. The presence of the terms "carnivore" and "dr" in both categories suggests that these films combine medical and influencer content kinds.

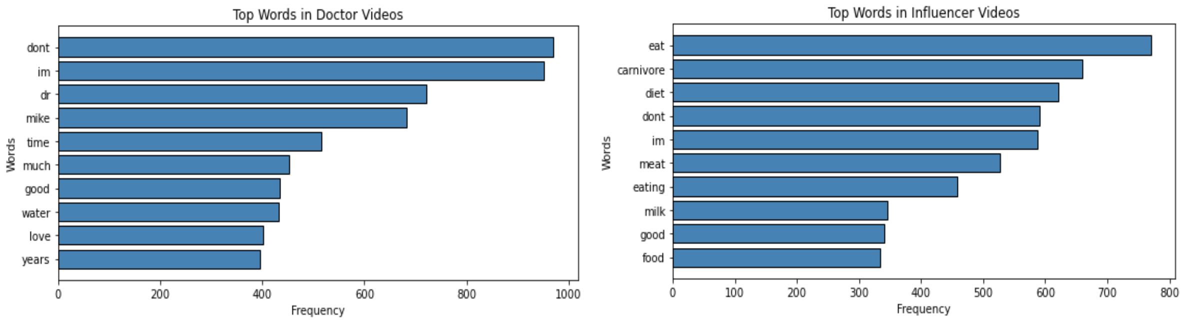

Figure 3: Unigram Analysis of Doctor and Influencer

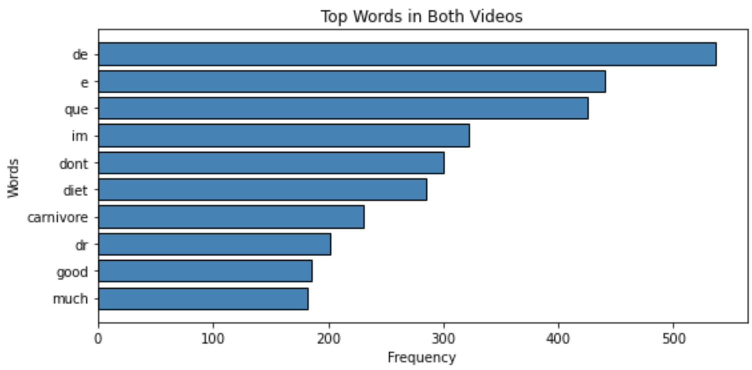

Figure 4: Unigram Analysis of Both Videos

4.2 Bigram Analysis

Bigram analysis was used to find significant two-word sentences that unigrams cannot capture which is visualised in Figures 5 and 6.

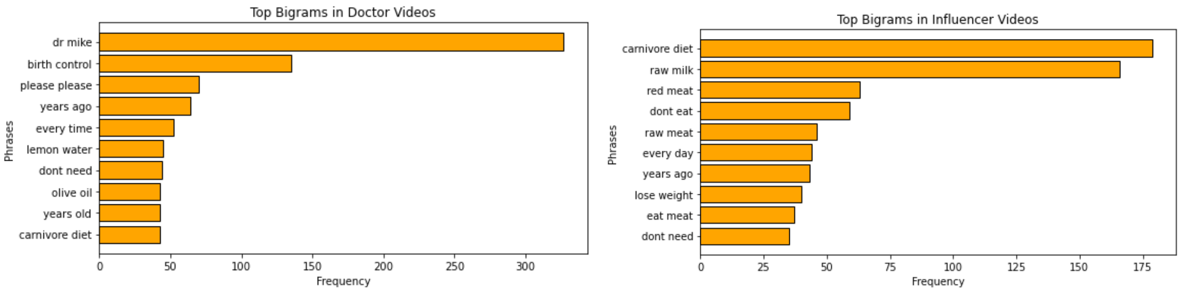

Figure 5: Bigram Analysis of Doctor and Influencer

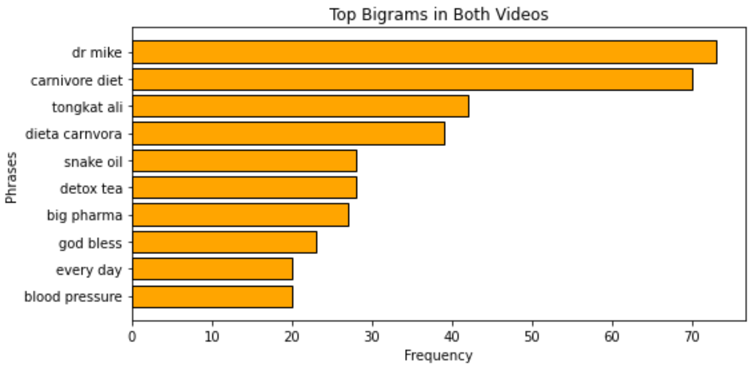

Figure 6: Bigram of Both Videos

The bigram "carnivore diet" appears frequently in both medical and influencer comment sections (43 and 180 instances, respectively), indicating that it is a popular topic of discussion across all content types. The influencer comments are heavily focused on specific food habits (raw milk, red meat), whereas doctor comments cover a wider range of medical and lifestyle themes (birth control, side effects, chest compressions).

4.3 Sentiment Analysis

Sentiment analysis was carried out using VADER (Valence Aware Dictionary and Sentiment Reasoner), a lexicon and rule-based sentiment analysis tool created exclusively for social media text. VADER assigns a composite sentiment score between -1 (most negative) and +1 (most positive). The comments were classed as good (score > 0.05), negative (score < -0.05), or neutral (otherwise). Tables 8 and 9 show the results, which are visualised in Figures 7 and 8.

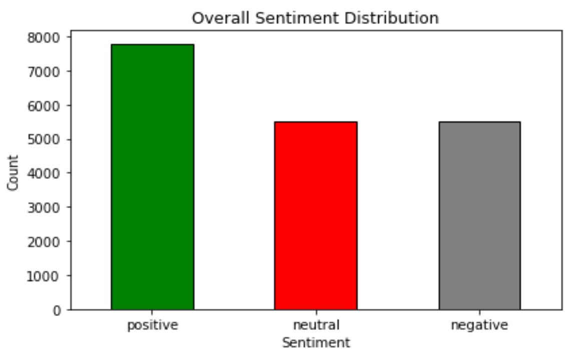

Figure 7: Sentiment Distribution of Comments

All three video categories received modestly favourable average sentiment scores, with just minor variations between them. Unlike predictions, influencer video comments were slightly more positive (0.0778) than doctor video comments (0.0721). This shows that audiences for both content categories prefer generally supportive or neutral opinion over blatantly hostile debate. The 'both' category had the greatest average sentiment (0.1004) as shown in Figure 8, presumably indicating a broader appreciation for work that connects both perspectives.

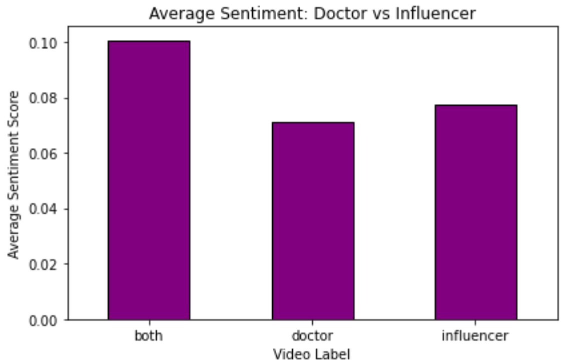

Figure 8: Average Sentiment Distribution of Doctors and Influencer

As shown in Table Figure 7, the overall distribution 41.6% positive, 29.3% negative, and 29.1% neutral, reflects a typical YouTube comment pattern in which positive engagement dominates but a significant minority of negative commentary exists, most likely representing sceptics and critics in both communities.

4.4 Topic Modelling (LDA)

Latent Dirichlet Allocation (LDA) was performed with scikit-learn's LatentDirichletAllocation.LatentDirichletAllocation with CountVectorizer (max_df=0.9, min_df=10). Five themes were retrieved from the entire dataset, with three extracted from each label subset for finer-grained analysis.

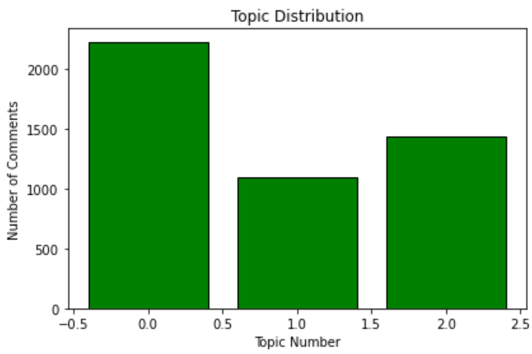

Figure 9: Distribution of Topics(LDA)

Figure 9 demonstrate that the topic model successfully distinguishes between food conversations (Topic 4, Influencer Topics 0 and 1) and general medical opinion (Doctor Topics 0 and 1). The presence of a multilingual topic (Topic 0) reflects YouTube's international audience for health material and demonstrates a weakness of the English-only filtering strategy. Doctor video comments are more thematically diverse, spanning from medical viewpoints to humour and particular health questions, whereas influencer video comments are more firmly focused on carnivorous diet content, implying a more ideologically orientated audience.

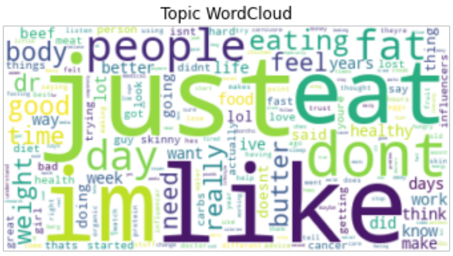

Figure 10: Word Cloud of User Comments

Figure 10 depicts a word cloud of the most common terms from all LDA subjects. The visualisation supports the dominance of diet and lifestyle jargon - phrases like "eat", "fat", "butter", "healthy", and "dont" are most common, indicating the carnivorous diet discourse that pervades both doctor and influencer comment sections. Words like "people", "im", and "just" show the conversational and intimate aspect of YouTube comment discussion. The word cloud supplements the LDA topic tables by offering a quick visual summary of the most important language in the sample.

5. Network Analysis

A directed interaction network was constructed from the YouTube comment reply data. Network properties are summarised in Table 4, and the degree distribution is shown in Figure 11. The network was defined as follows:

Table 4: Network Construction Properties and Rationale

Property	Definition	Rationale
Nodes	YouTube users (commenters)	Each unique author username becomes a node
Edges	Reply relationships (author → reply_to)	A directed edge from A to B means user A replied to user B
Direction	Directed	Reply direction captures who initiates interaction with whom
Weight	Unweighted	The existence of interaction is the focus, not frequency
Filtering	Top active users used for visualisation subgraph	Full graph has 2,206 nodes; visualisation uses top 50 by degree for readability

The degree distribution of the user interaction network is heavily skewed, with the vast majority of users having very low degree values (0-2 connections). This suggests that most users do not engage in reply-based exchanges and instead publish their own comments.

Table 5: Summary of Network Metrics

Network Metric	Value
Total nodes (users)	2,206
Total edges (reply interactions)	1,816

Network type	Directed (DiGraph)
Converted to undirected for centrality	Yes (G_co)

Only a tiny number of users have higher degree values, implying that very few people participate in multiple interactions or are more active contributors to the network.

5.2 Centrality Analysis

Four centrality measures were calculated on the undirected network to provide a comprehensive picture of user importance and interaction patterns. Figure 11 displays the top ten users in terms of degree centrality.

Degree Centrality

Degree centrality estimates the proportion of all other users to whom a given user is directly connected. A high degree of centrality suggests that the user participates in a large number of reply exchanges, either by responding to others or being responded to regularly.

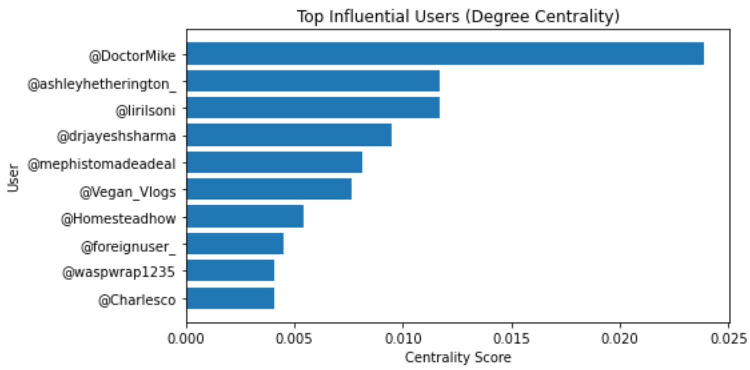

Figure 11: Top 10 Users by Centrality Score

According to Figure 11, the most influential user by degree centrality is @DoctorMike, a well-known medical YouTuber who has sparked numerous reply threads in his own comment sections. This is to be expected, as channel owners and highly visible public people inevitably accumulate reply engagements. The inclusion of @drjayeshsharma and @Vegan_Vlogs in the top ten implies that other content creators are actively participating in comment debates.

Betweenness Centrality

Betweenness centrality determines how frequently a user is on the shortest path between two other users in the network. Users with strong betweenness centrality serve as bridges between various conversation groups, playing an important role in integrating otherwise disparate communities. Given the relatively sparse reply network (1,816 edges spanning 2,206 nodes), most users have near-zero betweenness, implying that the network is primarily composed of isolated tiny discussion threads rather than a highly interconnected discussion web.

Eigenvector Centrality

Eigenvector centrality evaluates impact not only based on a user's own connections, but also the importance of the persons to whom they are connected. A user with fewer but more influential connections can outperform a user with many low-influence connections. This metric aids in the identification of users who are central to influential discussions rather than peripheral topics.

Clustering Coefficient

The clustering coefficient assesses the degree to which users tend to cluster together that is, whether a user's connections are also linked to one another. A high clustering coefficient in a network subregion implies close-knit discussion groups in which the same users communicate with one another on a regular basis, which could indicate echo chambers.

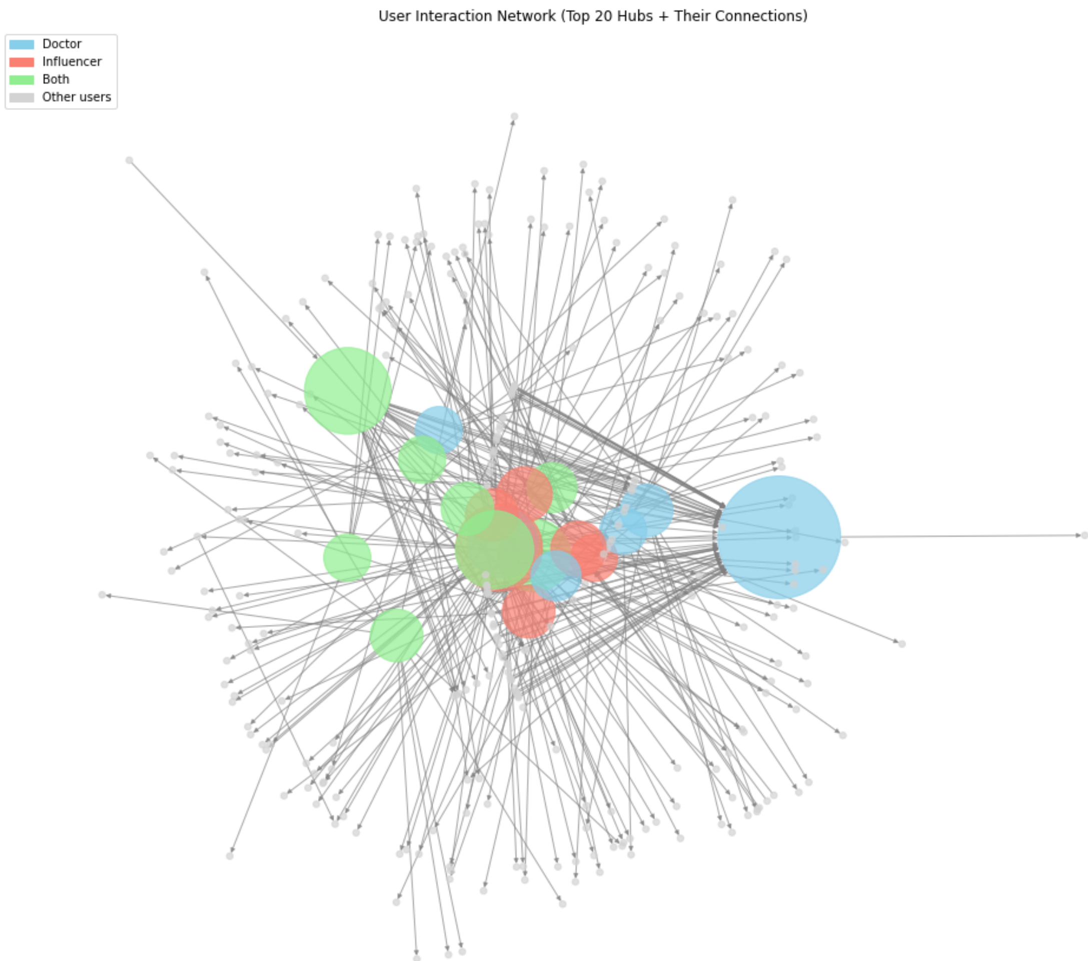

Figure 12: User Interaction Network Highlighting Top Hubs and Their Connections

5.3 Community Detection

The Louvain algorithm was used to detect communities, with the findings reported in Table 6 and the network visualised in Figure 13, which was constructed using the python-louvain module. The Louvain method is a modularity-optimization algorithm that iteratively merges nodes into communities to maximum modularity score. It was used on the undirected network.

Table 6: Summary of Community Detection Results

Community Metric	Value

Total communities detected	683
Modularity score	0.9936
Largest community	56 users (Community 206)
Second largest	53 users (Community 122)
Third largest	36 users (Community 507)
Majority of communities	Very small (2-5 users)

Interpretation of Community Structure

As indicated in Table 6, the modularity score of 0.9936 is extraordinarily high (the theoretical maximum is 1.0), indicating that the Louvain algorithm discovered very well-separated communities with dense internal linkages and sparse cross-community connections. This is congruent with the network structure: YouTube comment reply threads are naturally self-contained (a user responds within a specific video's comment section), resulting in several small, isolated clusters.

The detection of 683 communities from 2,206 nodes implies that the average community size is roughly 3.2 users, indicating that the majority of response interactions are short back-and-forth exchanges between 2-4 users within a single video's comment area rather than vast cross-video discussion networks.

Figure 13: User Interaction Network Coloured by Detected Communities (Louvain Method)

The user interaction network was visualised using Louvain community detection, where each colour represents a distinct community of users. As shown in Figure 13, the graph reveals that the network is divided into multiple small clusters rather than a few large, well-defined communities.

Most communities appear loosely connected, with only a limited number of edges linking different groups. This indicates that user interactions are fragmented, and discussions tend to occur in small, isolated clusters rather than forming large, cohesive communities.

The presence of several disconnected or weakly connected nodes further highlights the sparse nature of the network, suggesting that users primarily engage independently rather than participating in sustained group discussions.

Content Alignment of Communities

Analysis of the popular video label within each group revealed a clear pattern: content type is nearly entirely uniform. The great majority of discovered groups included users who communicated only in doctor-labeled video comment parts or influencer-labeled video comment sections. Only a few communities bridged both content types.

This study lends support to the concept that YouTube comment communities form around certain content genres, with little cross-pollination between medical and influencer viewers.

Users who comment on doctor debunking videos communicate largely with other users in the video's comment box, and the same is true for influencer videos. This points to potential audience fragmentation and limited cross-community discourse about health information on YouTube.

7. Conclusion

This project used a complete social media and network analytic pipeline to analyse YouTube comment data, comparing the comment behaviour, language, mood, and community structure of audiences that engaged with medical professional debunking video against wellness influencer content. Using a dataset of 89 videos and 18,447 cleaned and deduplicated comments collected via the YouTube Data API v3, the analysis combined NLP techniques (tokenisation, word frequency analysis, bigram analysis, VADER sentiment analysis, and LDA topic modelling) with network analysis (directed reply graph construction, degree/betweenness/eigenvector centrality, clustering coefficient, and Louvain community detection).

The results provide a nuanced response to the research topic. Licensed medical professionals have stronger network authority @DoctorMike and other medical accounts lead degree centrality rankings, and doctor videos receive more than twice as many comments as influencer videos (10,458 vs 4,756). However, wellness influencers wield ideological authority in their communities: influencer comment sections use highly concentrated, ideologically unified jargon (carnivore diet, raw milk, meat), implying a loyal and focused audience.

Sentiment scores are roughly comparable (0.072 vs 0.078), indicating that neither content type elicits stronger emotional approval. The reply network is typified by numerous small, isolated communities with very high modularity (0.9936), and communities are well linked with content type, showing that doctor and influencer audiences are highly separated and do not interact significantly online.

These findings suggest that, while licensed medical professionals have greater network-level authority on YouTube, wellness influencers maintain strong ideological authority within their own communities and because these two audiences are nearly completely separated, the two forms of authority rarely clash directly online. This has significant implications for public health communication: engaging influencer audiences may necessitate techniques that work within their current community structures rather than depending on cross-community exposure to medical authority.